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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 30: Sustainable Transport Planning and Approaches-I

The Traditional Transport Planning Process

Prof. Bhola Ram Gurjar

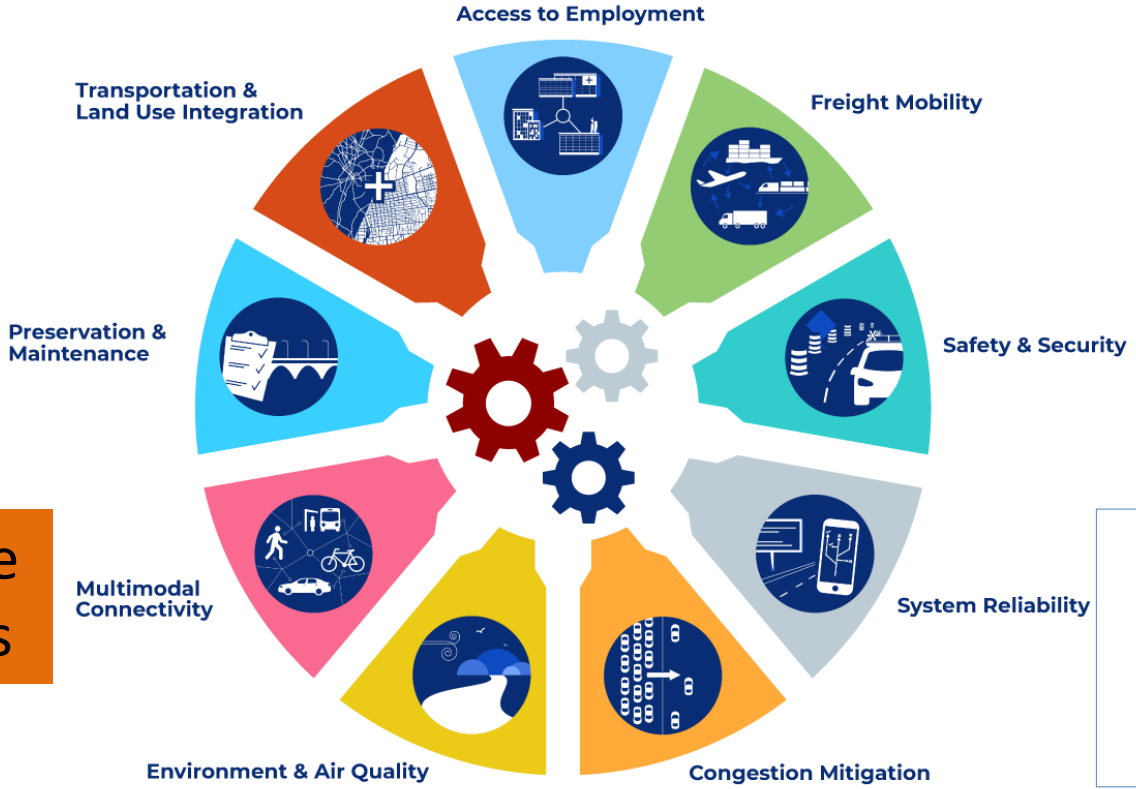
Dept. of Civil Engineering



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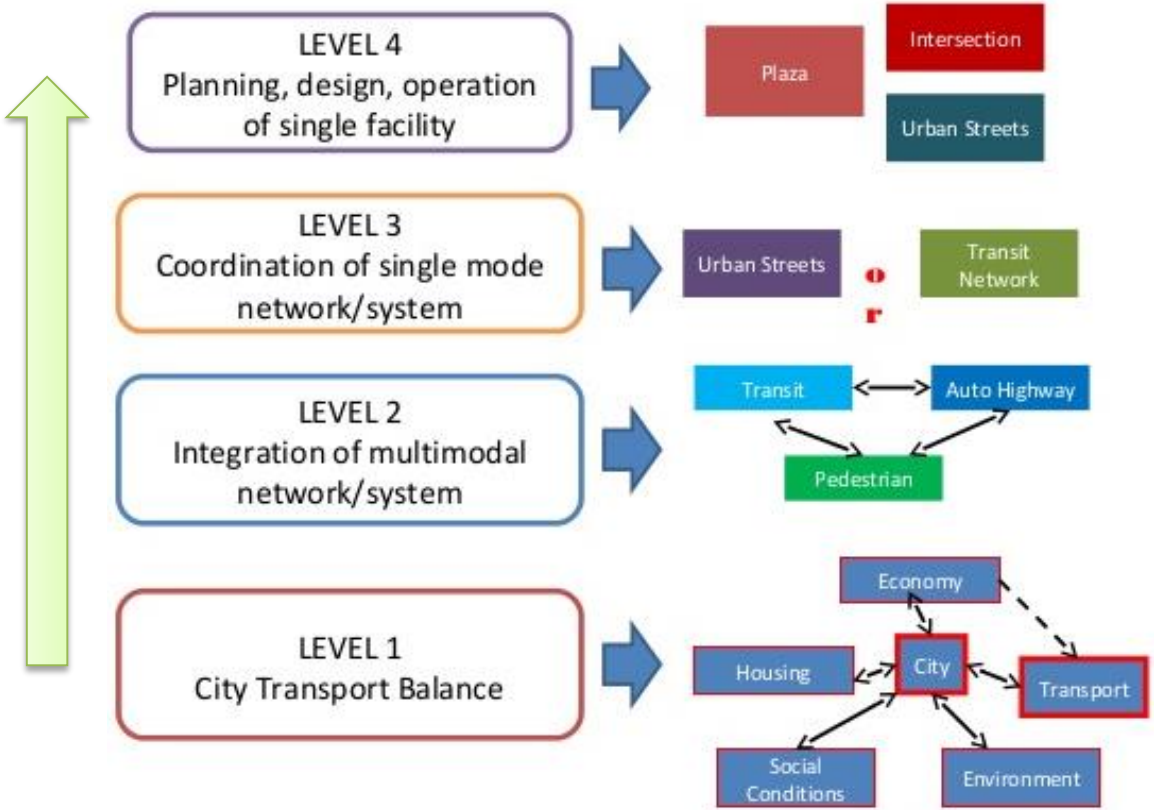
Goals of Transport Planning



These are the Goals

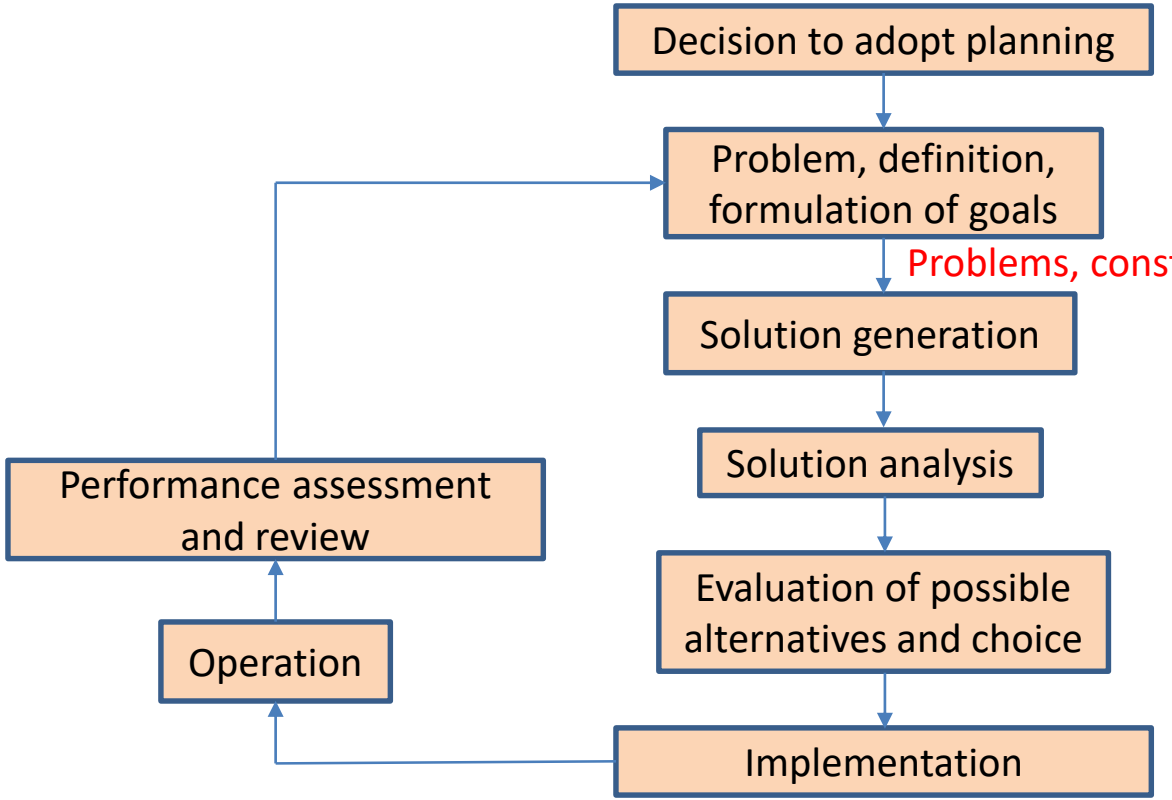
Source: (PlanRva, 2019)

Planning future Transportation



Source: (Piter Biswas, 2016, https://www.slideshare.net/PiterBiswas/urban-transportation-planning-66010727)

Systems Approach to Transport Planning



Source: (L. R. Kadiyali 2017)

The Traditional 4-Step Planning Model

Step 1. Trip Generation

Forecasts the travel demands for a specified area- number of trips, purpose etc. in a study area

Step 2. Trip Distribution

Forecasts the number of trips between Origin and Destinations in each zones of study area

Step 3. Modal Split (Mode choice)

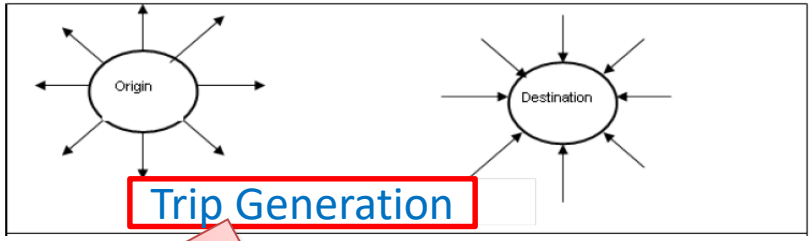
Predicts the mode choice preference of individuals

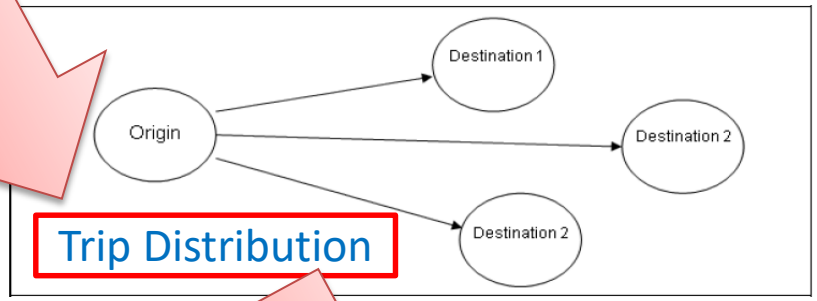
Step 4. Trip Assignment

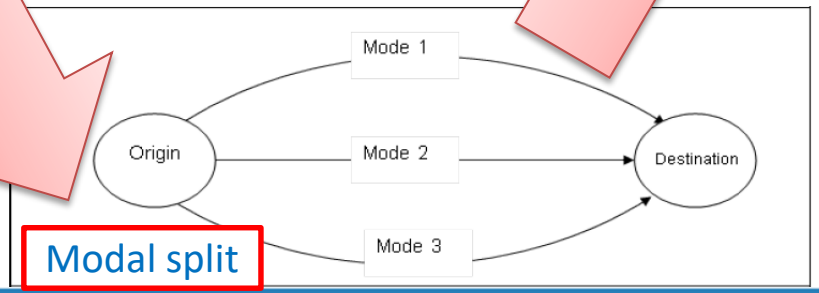
Assigns traffic to a predicted model

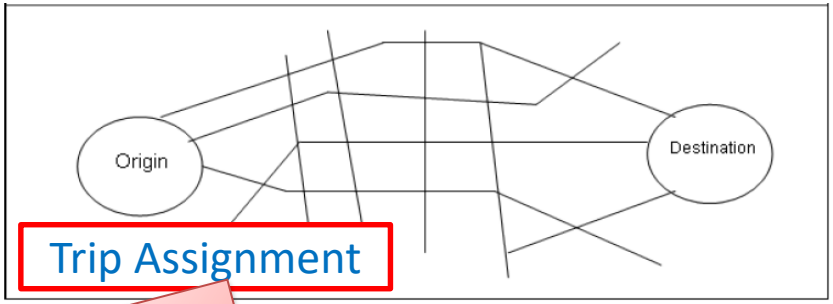
Source: (L. R. Kadiyali 2017)

Key Steps in the Traditional 4-Step Planning Model









What we forecast in a Planning process?

- How many total trips ?
- How long is an average trip between Origin and destinations?
- What is the Modal split (Mode choice)?
- How many cars per hour per lane?
- How to provide Public transport (number of trips, frequency etc.)?

Stages in Transport Planning

- Survey and analysis of existing conditions
- Forecast, analysis of future conditions and plan synthesis
- Evaluation
- Programme adoption and implementation
- Continuing studies

Step 1. Trip Generation
Forecasts the travel demands for a specified area- number of trips, purpose etc. in a study area

Step 2. Trip Distribution
Forecasts the number of trips between Origin and Destinations in each zones of study area

Step 3. Modal Split (Mode choice)
Predicts the mode choice preference of individuals

Step 4. Trip Assignment
Assigns traffic to a predicted model

The Traditional Transport forecasting process are completed by using these 5 stages of planning.

Source: (L. R. Kadiyali 2017)

Stages in Transport Planning

Stage 1: Survey and analysis of existing conditions

Definition of Survey area and formulation of Goals

Inventory of existing travel pattern
- Origin and destination surveys
- Household interviews
- Rail and bus surveys
- Goods vehicle interviews
- Coach surveys
- Cordon and screen line studies

Inventory of existing transport facilities
- Street inventory
- Travel time studies
- Bus inventory
- Rail inventory
- Parking inventory

Land use and economic activities
- Zoning
- Land use
- Population structure
- Household structure
- Employment pattern
- Income

Stage 1

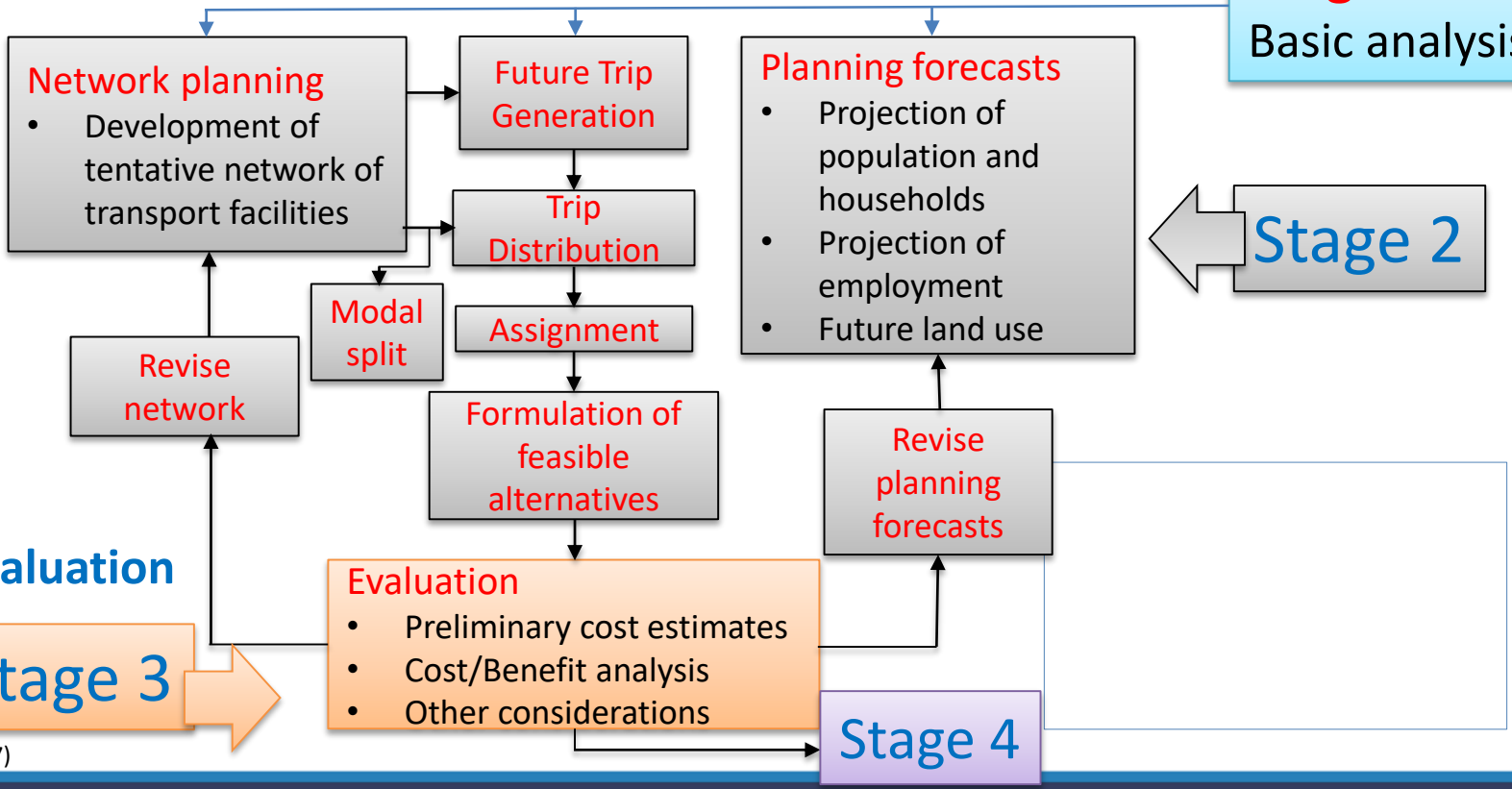
Basic Analysis
- Car ownership
- Trip generation
- Trip attraction
- Modal selection
- Trip distribution
- Goods vehicles trips

Stage 2

Source: (L. R. Kadiyali 2017)

Stages in Transport Planning

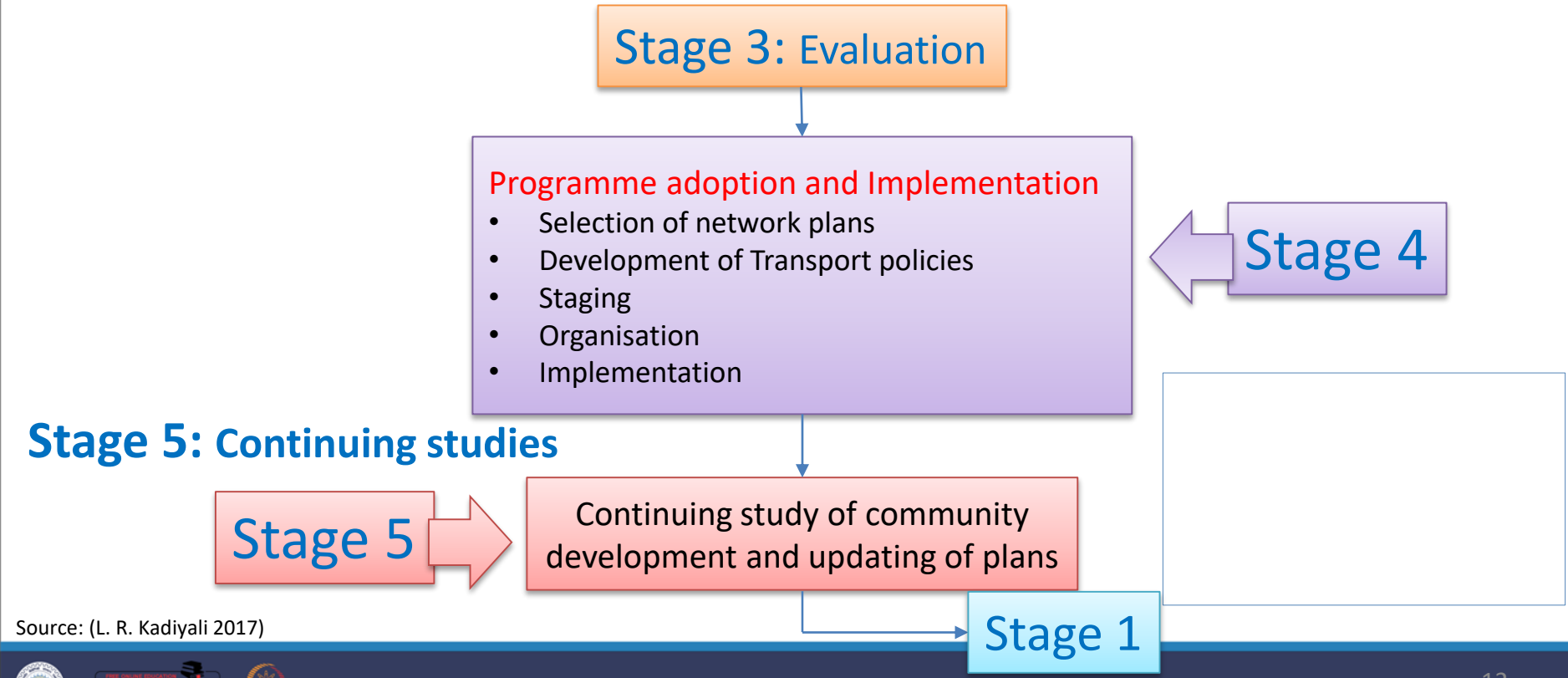
Stage 2: Forecast, analysis of future conditions and plan synthesis



Source: (L. R. Kadiyali 2017)

Stages in Transport Planning

Stage 4: Programme adoption and implementation



Source: (L. R. Kadiyali 2017)

Transportation Planning: Factors to Consider



Factors to consider

Source: (PlanRva, 2019)

Transportation Surveys

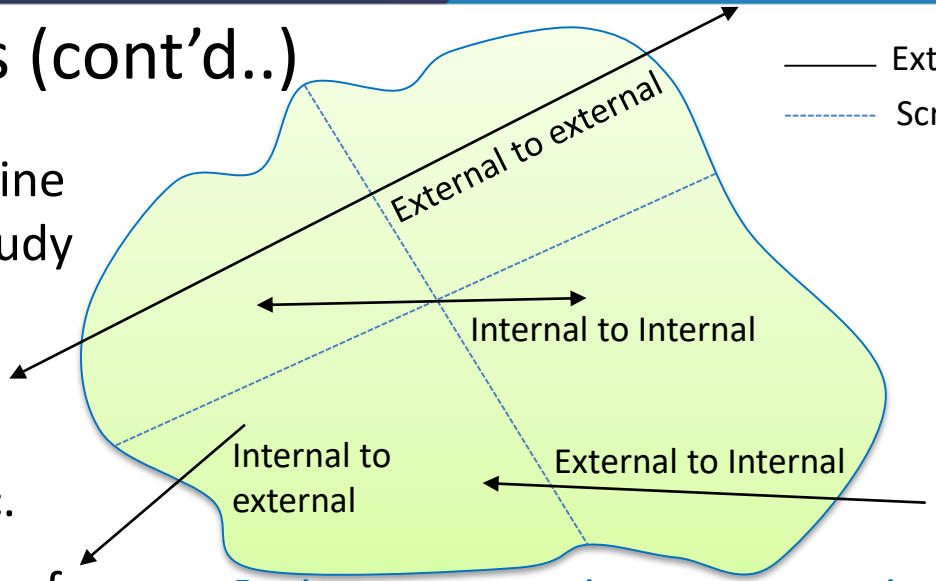


- First stage in the formulation of a Transportation plan.
- Aim is to collect data on all factors influencing travel patterns.
- Voluminous and time consuming work (Can take upto 2 years)
- Cost involved is very high.

Source: (L. R. Kadiyali 2017)

Transportation Surveys (cont'd..)

- External cordon: Imaginary line representing boundary of study area.
- Screen line: located along physical or natural barriers
  - Ex. Rivers, railway lines etc.

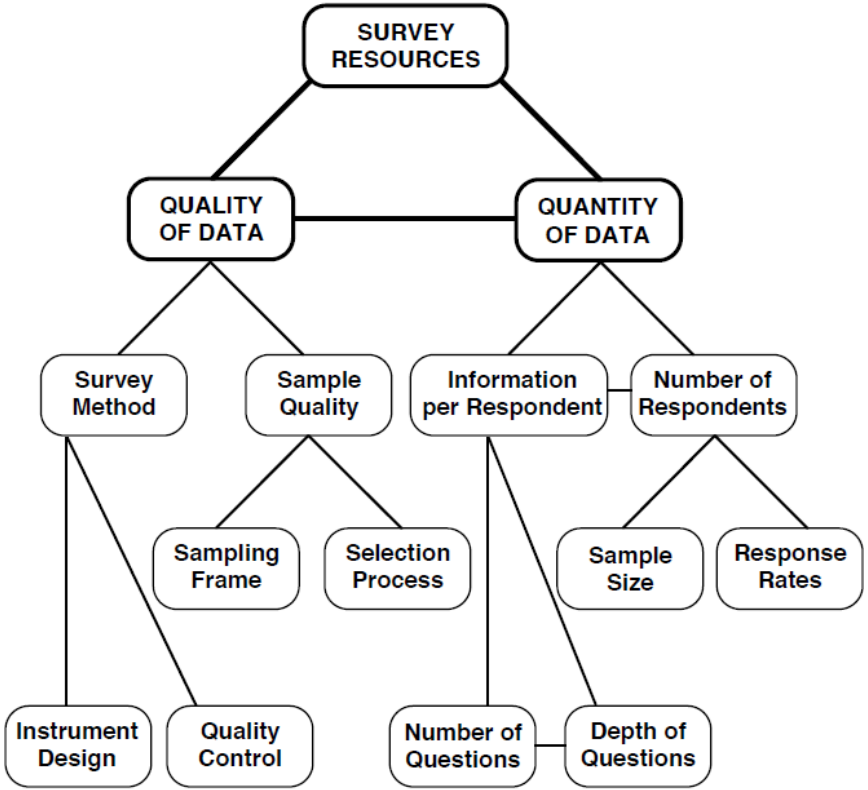


Basic movements in a transportation survey

- 4 Basic movements in survey of a study area
  - (i) Internal to internal
  - (ii) Internal to external
  - (iii) External to internal
  - (iv) External to external

Source: (L. R. Kadiyali 2017)

Transportation Surveys: Trade-Offs





Source: (Anthony J. Richardson et. al, 1995)

Types of Transportation Surveys

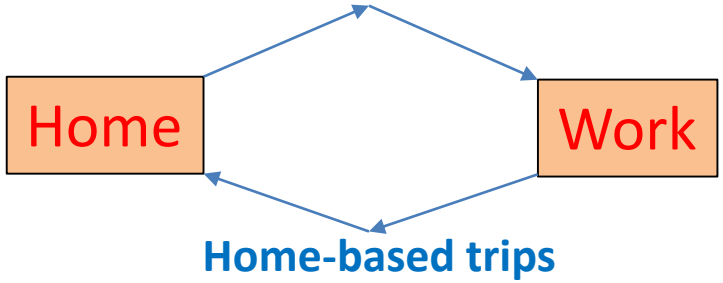


Source: (L. R. Kadiyali 2017)

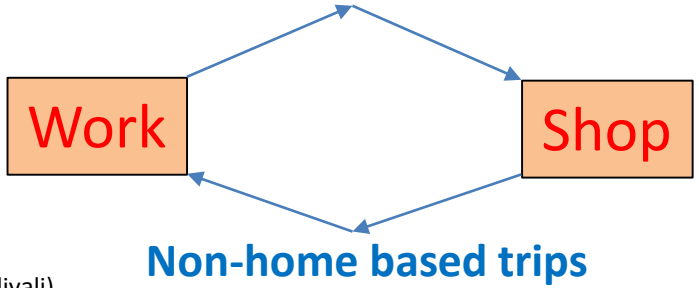
Trip Generation

- Aim is to calculate the number of trips in a given area

Trip Purpose



Home based trips constitutes nearly 80-90% of the total trips in a given area.



Source: (Traffic Engineering and Transport Planning, L. R. Kadiyali)

Factors governing Trip Generation and Attraction

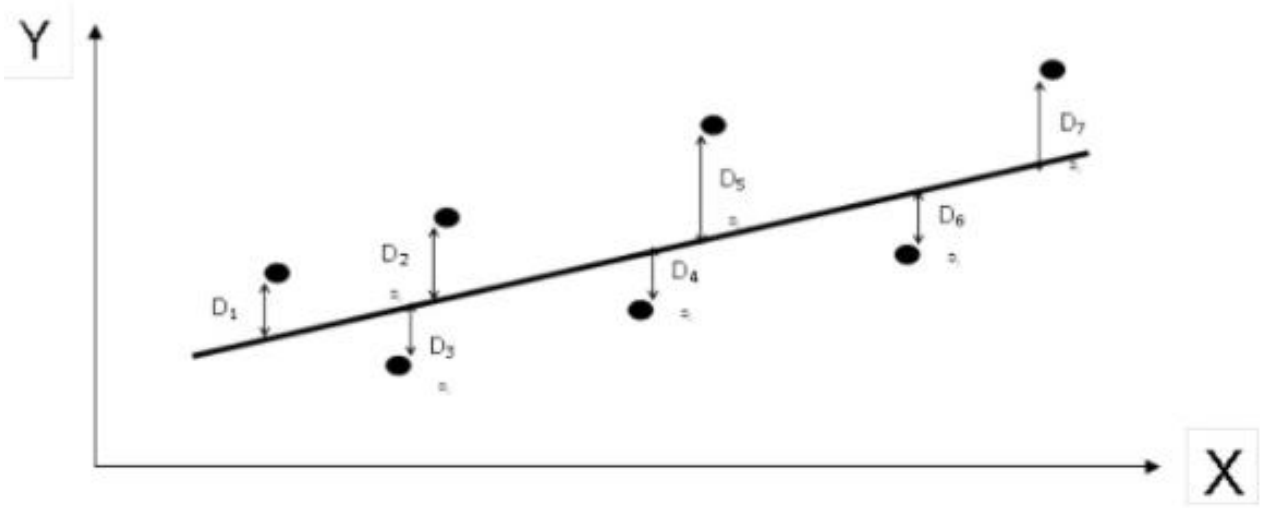
1. Income
2. Car ownership
3. Family size and composition
4. Land use characteristics
5. Distance of the zone from the town centre
6. Accessibility to public transport system and its efficiency
7. Employment opportunities



Source: (L. R. Kadiyali, 2017)

Trip Generation: Calculation

Multiple Linear Regression Analysis



The best line – the line that minimise $D_1 + D_2 + D_3 + \ldots + D_7$

Source: (Riza Atiq bin O.K. Rahmat , 2012)

Trip Generation: Calculation of Total Number of Trips

Multiple Linear Regression Analysis

Dependent variable: Number of trips

Independent variables: Various independent factors that influence trip generation

$$Y_p = a_1X_1 + a_2X_2 + a_3X_3,\ldots\ldots,a_nX_n + U$$

Where, Y_p = number of trips for specified purpose p

$X_1, x_2, x_3, \ldots, X_n$ = independent variables

$a_1, a_2, a_3 \ldots, a_n$ = coefficients of the respective independent variables

U = disturbance term, which is a constant, and representing that portion of Y_p not explained by the independent variables

Source: (L. R. Kadiyali, 2017)

To calculate the Total Number of Trips in an area

Numerical example 1

Dependent variable: Y = Average trips per dwelling unit

Independent variables: A = Car ownership, H = Household size, X_1 = Social rank index and X_2 = Urbanization index with coefficients 3.404, 0.516, 0.0119 and -0.343 respectively

Disturbance, U = 2.18

$$Y_p = a_1X_1 + a_2X_2 + a_3X_3, \ldots\ldots, a_nX_n + U$$

The Average number of trips per dwelling unit can be calculated by,

$$Y = 3.404\ A + 0.516\ H + 0.0119\ X1 - 0.343\ X2 + 2.18$$

Source: (L. R. Kadiyali, 2017)

Trip Rate

- Defined as the ratio of the total number of trips generated per day to the total population of considered area

$$Trip\ rate = \frac{Total\ number\ of\ trips\ generated\ per\ day}{Population\ of\ area\ considered}$$

Source: (L. R. Kadiyali 2017)

Calculation of Trip Rate

Numerical problem 1

Estimate trip rate for a residential colony with a population of 2744 and 6574 number of trips ?

Solution

$$Trip\ rate = \frac{Total\ number\ of\ trips\ generated\ per\ day}{Population\ of\ area\ considered}$$

$$Trip\ rate = \frac{6574}{2744} = 2.39 \approx 2.4$$

Source: (L. R. Kadiyali 2017)

Example of Trip Generation Calculation for Model development using Multiple Linear Regression Analysis

Step 2.

- Compute each of the parameters of the potential Regression equations.

Step 3.

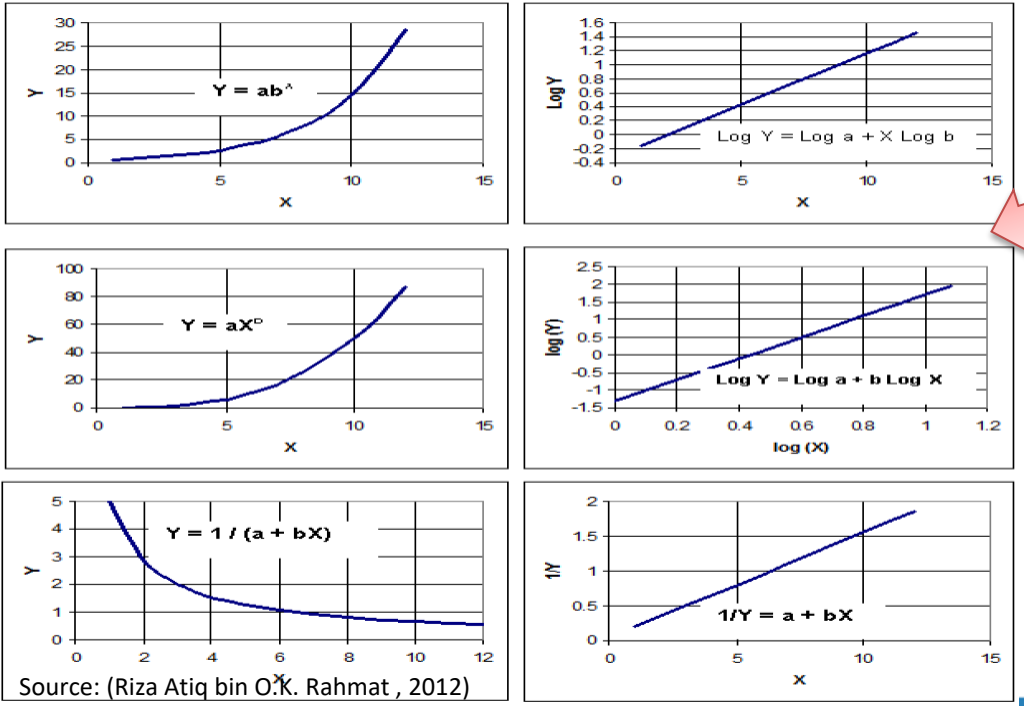
Check the following criteria for Statistical significance.

- R^2 value, Intercept value and Regression parameters

Source: (Riza Atiq bin O.K. Rahmat , 2012)

Example of Trip Generation Calculation for Model development using Multiple Linear Regression Analysis

Non-linear relationship could be linearised



Source: (Riza Atiq bin O.K. Rahmat , 2012)

To observe any relationship between parameters

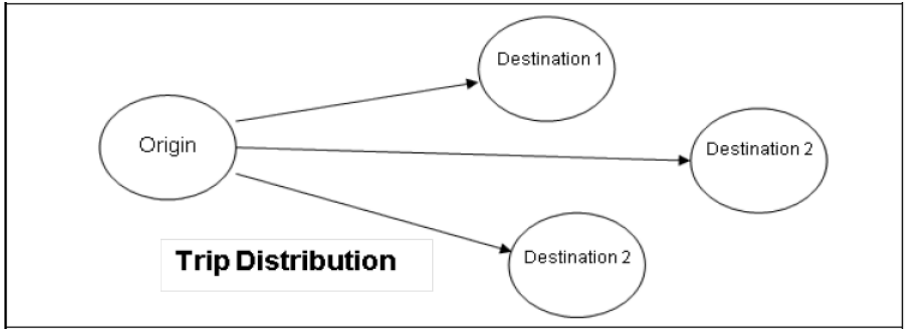
R^2 = 1 means Max. correlation between X and Y

R^2 = 0 means No correlation

Trip Distribution

Main Question addressed is

"Where will the generated trips go to?"



- Internal or External trips in a city
- Internal trips with Origin and Destination confined to city limits
- External trips whose Destination is beyond city or town limits

Source: (Riza Atiq bin O.K. Rahmat , 2012)

Main Purpose of Trip Distribution

Trip Production	Trip Attraction
560	1250
750	530
1105	430
545	540
450	1200
1040	500
4450	**4450**

Trips generated (origin) and attracted (destination) should be balanced and made equal

Source: (Riza Atiq bin O.K. Rahmat , 2012)

Balancing the O-D Matrix in Trip Distribution

	1	2	3	4	5	6	
1	157	67	54	68	151	63	560
2	211	89	72	91	202	84	750
3	310	132	107	134	298	124	1105
4	153	65	53	66	147	61	545
5	126	54	43	55	121	51	450
6	292	124	100	126	280	117	1050
	1250	530	430	540	1200	500	4450

Trip Production

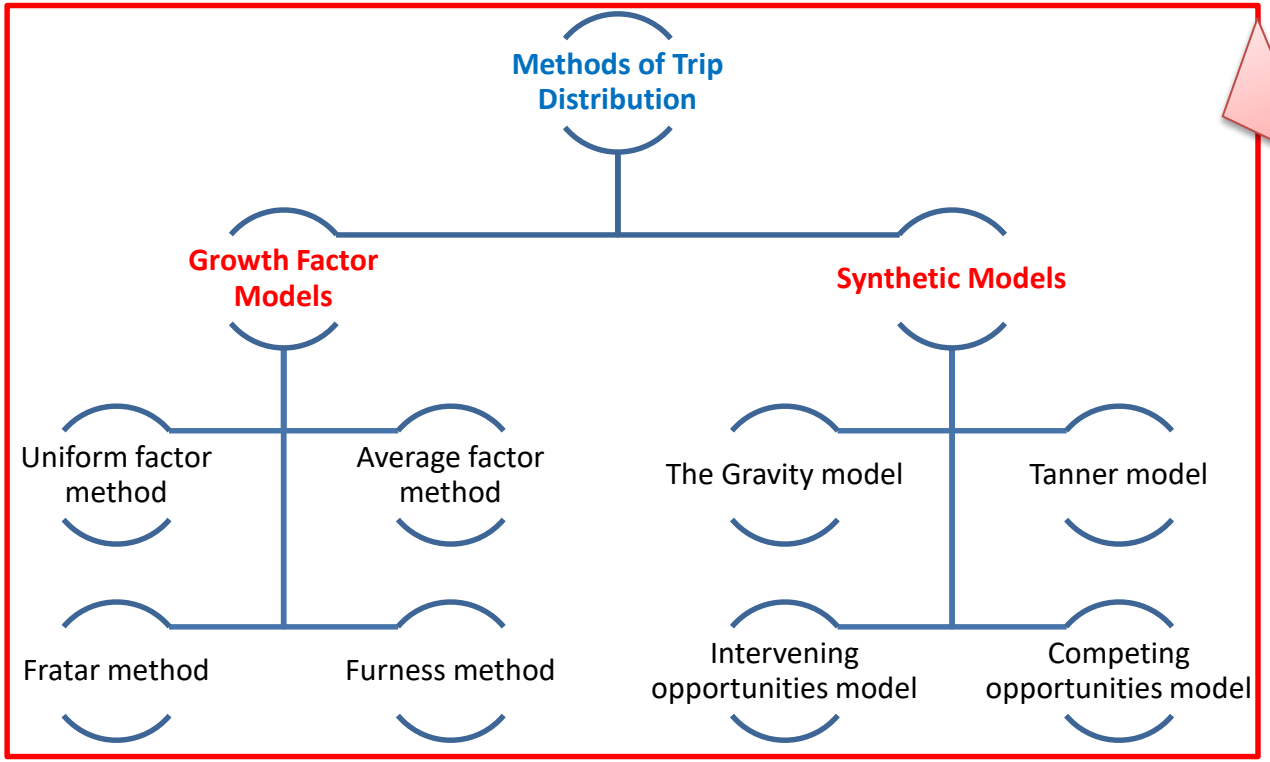
Trip Attraction

Trip Production	Trip Attraction
560	1250
750	530
1105	430
545	540
450	1200
1040	500
4450	4450

$$Example:\ 1250 \times \frac{1050}{4450} = 292$$

Source: (Riza Atiq bin O.K. Rahmat , 2012)

Methods of Trip Distribution

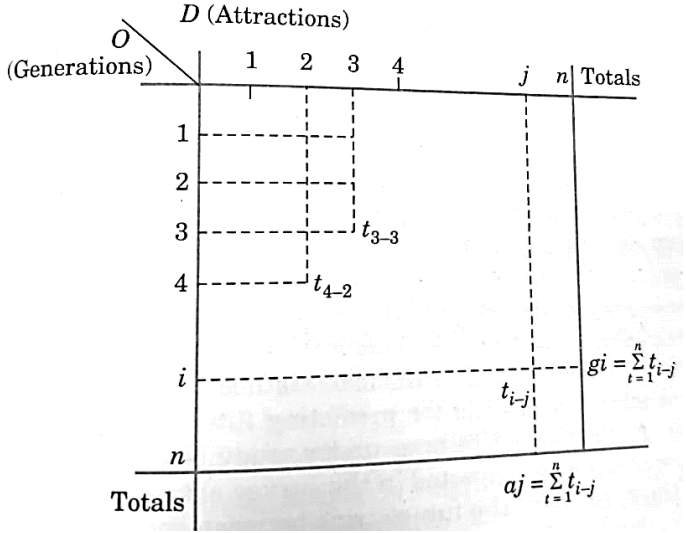


There are various methods to calculate the trip distribution for a study area

Source: (L. R. Kadiyali, 2017)

Growth Factor Models

- Assumes that we have a basic trip matrix data from previous studies or recent surveys.
- Main Goal here is to estimate future matrix.
- Used in earlier studies widely, but now been used only for smaller studies.
- Nowadays, Synthetic models are widely used.



Origin-Destination matrix

Source: (L. R. Kadiyali, 2017)

Example of a Uniform Growth factor model

Given a trip matrix for base year as in Table 1. If the traffic growth factor for the forecast year is 1.2 for the study area, estimate the future Trip matrix?

Table 1. Base year Trip matrix

	1	2	3	4	
1	157	67	54	68	560
2	211	89	72	91	750
3	310	132	107	134	1105
4	153	65	53	66	545
	1250	530	430	540	**4450**

Table 2. Estimated Future Trip matrix

	1	2	3	4	
1	188.4	80.4	64.8	81.6	672
2	253.2	106.8	86.4	109.2	900
3	372	158.4	128.4	160.8	1326
4	183.6	78	63.6	79.2	654
	1500	636	516	648	**5340**

Total Future trips increased from 4450 to 5340 trips

$$Future\ trip = Base\ year\ trip \times \text{Growth factor}$$

Growth factor from past data

$$Example: 157 \times 1.2 = 188.4$$

Source: (L. R. Kadiyali, 2017)

Synthetic model (The Gravity model)

- Widely used in Current studies.

- Based on Newton's law of gravitation
  - "The force of attraction between two bodies is directly proportional to the product of masses between the two bodies and inversely proportional to the square of the distance."

- The Gravity model is the well known synthetic model used widely.

$$F = G\frac{m_1 m_2}{D^2}$$

$where,$
$m1\ and\ m2 =$ masses of bodies
D = distance between the bodies

Source: (L. R. Kadiyali, 2017)

The Gravity model

Newton's law of gravitation → $F = G\frac{m_1 m_2}{D^2}$

Total Number of trips between origin and destination → $T_{ij} = K\frac{P_i A_j}{f(Rij)}$

where,

T_{ij} = Number of trips between origin I and destination j.

P_i = Production of zone I

A_j = Attraction of zone j

$f(Rij)$ = Separation function of the travel cost between zone i and zone j

Separation function depends on distance and mode

Source: (L. R. Kadiyali, 2017)

Modal Split/ Mode Choice

- The process of separating person-trips by the mode of travel (Choice of people)

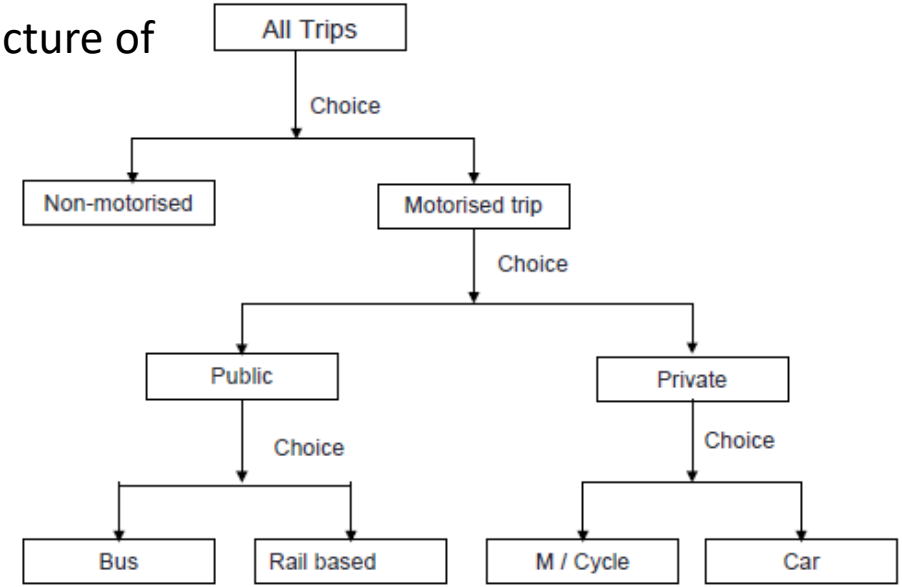








Decision structure of All Trips



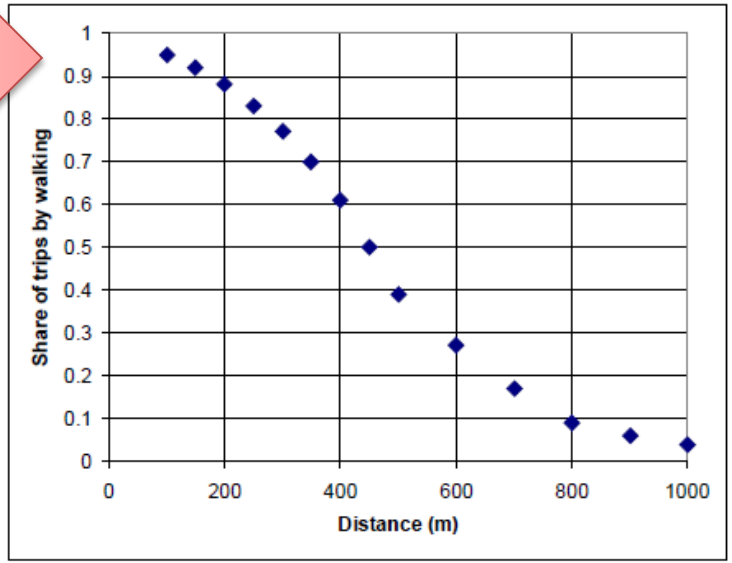
Source: (L. R. Kadiyali, 2017; Riza Atiq bin O.K. Rahmat , 2012)

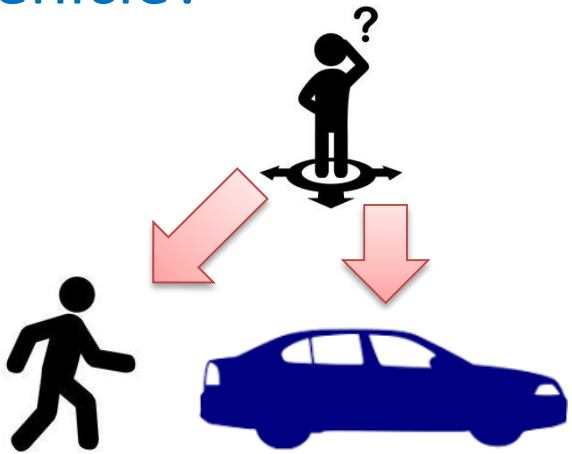
Example: Modelling the Choice of people

To Choose between Walking or to Ride a vehicle?

Distance (m)	Share of trips by walking
100	0.95
150	0.92
200	0.88
250	0.83
300	0.77
350	0.70
400	0.61
450	0.5
500	0.39
600	0.27
700	0.17
800	0.09
900	0.06
1000	0.04

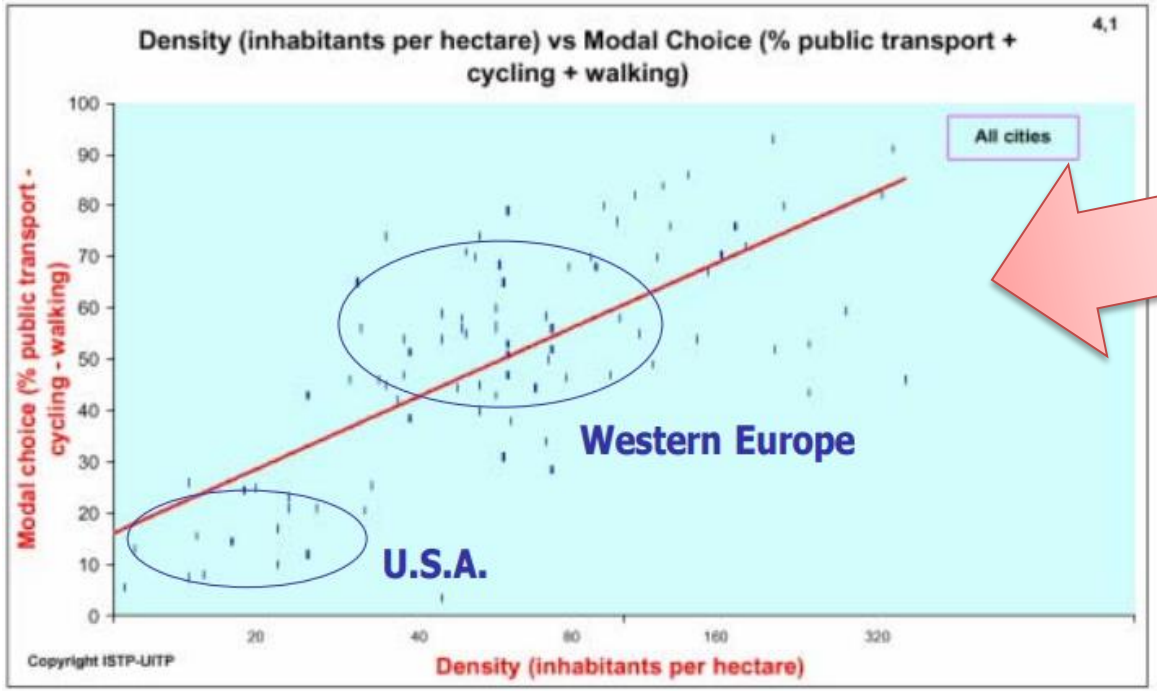
Example of Plotting the share of trips by Walking





Source: (Riza Atiq bin O.K. Rahmat , 2012)

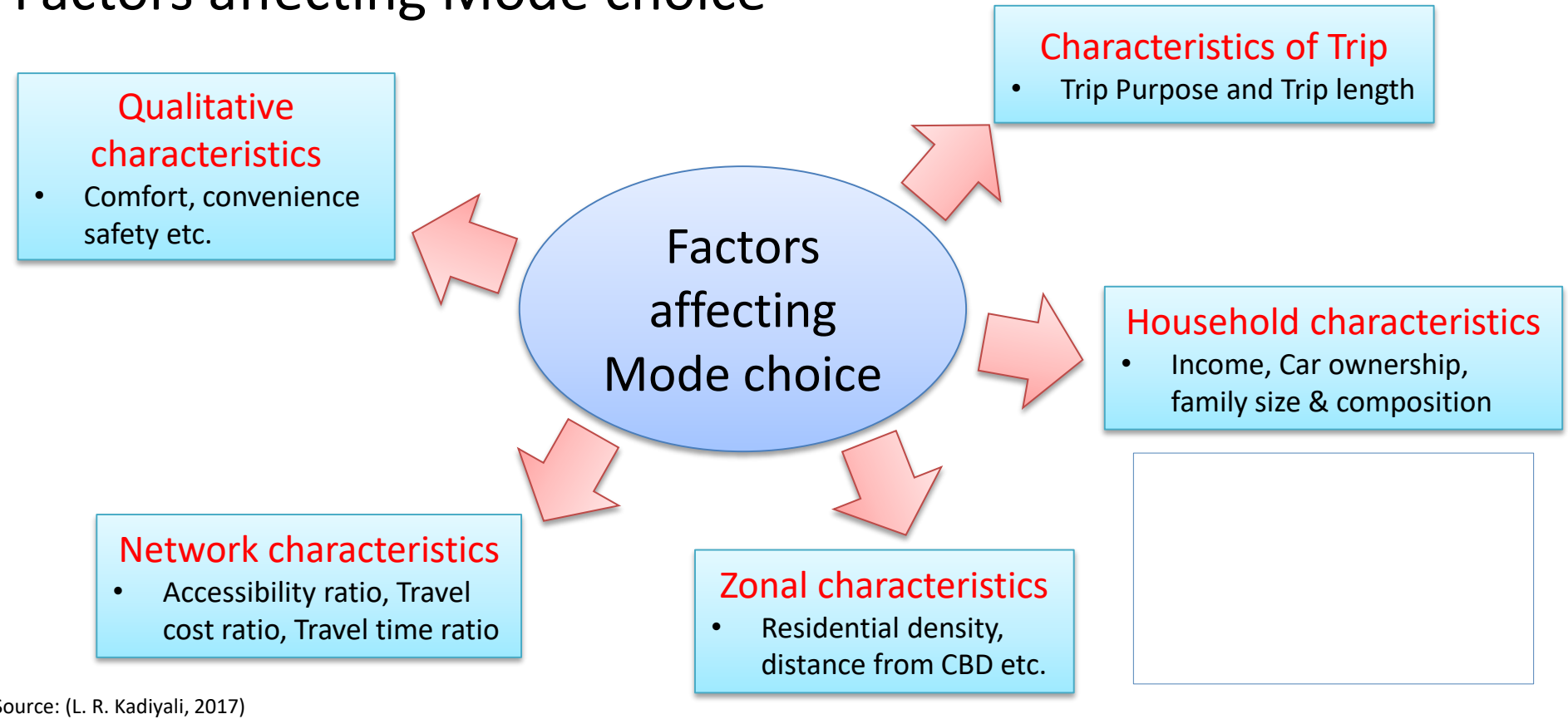
Example of Mode choice of People in the U.S and Western Europe



People of Western Europe prefer Public transport, Cycling or walking compared to the Private vehicles

Source: (MIT lecture Notes, 2002)

Factors affecting Mode choice



Source: (L. R. Kadiyali, 2017)

Modal Split: Calculation

- Measures the satisfaction derived from a mode choice

- Expressed as linear weighted sum of independent variables

$$U = a_0 + a_1X_1 + a_2X_2 + a_3X_3,\ldots\ldots,a_nX_n$$

Where, U = utility derived from choice
X_n = attributes
a_n = model parameters

Source: (L. R. Kadiyali, 2017)

Modal Split: Calculation by Binary Logit Model

- Calculates the probability of selecting a particular mode of transport

$$p(K) = \frac{e^{Uk}}{\sum e^{Uk}}$$

Where, p = probability of selecting mode k

The probability of choosing public transport (pt) over automobile (auto) between zones i and j ,

$$p_{ij} = \frac{e^{Upt}}{e^{Upt} + e^{Uauto}}$$

The probability of choosing automobile (auto) over public transport (pt) between zones i and j ,

$$p_{ij} = \frac{e^{Uauto}}{e^{Upt} + e^{Uauto}}$$

Source: (L. R. Kadiyali, 2017)

Modal Split: Numerical example 1

Qn: Suppose, the utility function of public transport (pt) and automobile (auto) are represented by the utility equation:

$$U = a_k - 0.35t_1 - 0.08t_2 - 0.005c$$

Where, a_k = mode specific variable

t_1 = travel time in minutes

t_2 = waiting time in minutes

c = cost (cents)

and the travel characteristics between two zones are given in table below:

Variable	Auto	Public transport
a_k	-0.46	0.47
t_1	20	30
t_2	8	6
c	320	100

Calculate the probability of choosing automobile and public transport w.r.t other mode.

Source: (Reginald Souleyrette, and Adam Kirk, 2013)

Modal Split: Numerical example 1

Solution:

$$U_{auto} = -0.46 - 0.35(20) - 0.08(8) - 0.005(320) = -9.70$$

$$U_{pt} = -0.07 - 0.35(30) - 0.08(6) - 0.005(100) = -11.55$$

$$p_{auto} = \frac{e^{Uauto}}{e^{Upt} + e^{Uauto}} = \frac{e^{-9.70}}{e^{-9.70} + e^{-11.55}} = 0.86 \; or \; 86\%$$

$$p_{pt} = \frac{e^{Upt}}{e^{Upt} + e^{Uauto}} = \frac{e^{-11.55}}{e^{-9.70} + e^{-11.55}} = 0.14 \; or \; 14\%$$

Source: (Reginald Souleyrette, and Adam Kirk, 2013)

Modal Split: Numerical example 2

Qn: Suppose, the utility function of public transport (pt) and automobile (auto) are represented by the utility equation:

$$U = a_k - 0.35t_1 - 0.08t_2 - 0.005c$$

Where, a_k = mode specific variable
t_1 = travel time in minutes
t_2 = waiting time in minutes
c = cost (cents)

and the travel characteristics between two zones are given in table below:

Variable	Auto	Public transport	Bike
a_k	-0.46	-0.07	-0.07
t_1	20	30	35
t_2	8	6	0
c	320	100	0

Calculate the probability of choosing each modes w.r.t other mode.

Source: (Reginald Souleyrette, and Adam Kirk, 2013)

Modal Split: Numerical example 2

Solution:

$U_{auto} = -0.46 - 0.35(20) - 0.08(8) - 0.005(320) = -9.70$

$U_{pt} = -0.07 - 0.35(30) - 0.08(6) - 0.005(100) = -11.55$

$U_{bike} = -0.07 - 0.35(35) - 0.08(0) - 0.005(0) = -12.32$

$$p_{auto} = \frac{e^{Uauto}}{e^{Upt}+e^{Uauto}+e^{Ubike}} = \frac{e^{-9.70}}{e^{-9.70}+e^{-11.55}+e^{-12.32}} = 0.81 \; or \; 81\%$$

$$p_{pt} = \frac{e^{Upt}}{e^{Upt}+e^{Uauto}+e^{Ubike}} = \frac{e^{-11.55}}{e^{-9.70}+e^{-11.55}+e^{-12.32}} = 0.13 \; or \; 13\%$$

$$p_{bike} = \frac{e^{Ubike}}{e^{Upt}+e^{Uauto}+e^{Ubike}} = \frac{e^{-12.32}}{e^{-9.70}+e^{-11.55}+e^{-12.32}} = 0.06 \; or \; 6\%$$

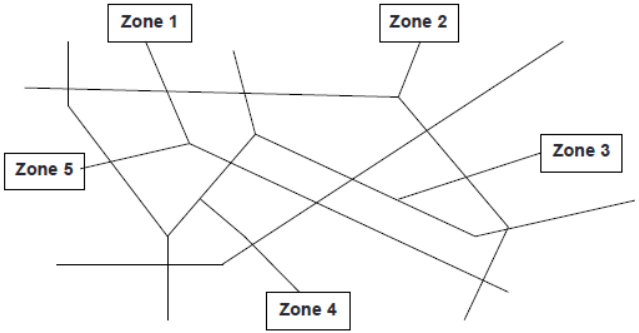
Inference:

The utility of automobile stays the same even though its share of passengers reduced

Source: (Reginald Souleyrette, and Adam Kirk, 2013)

Trip Assignment

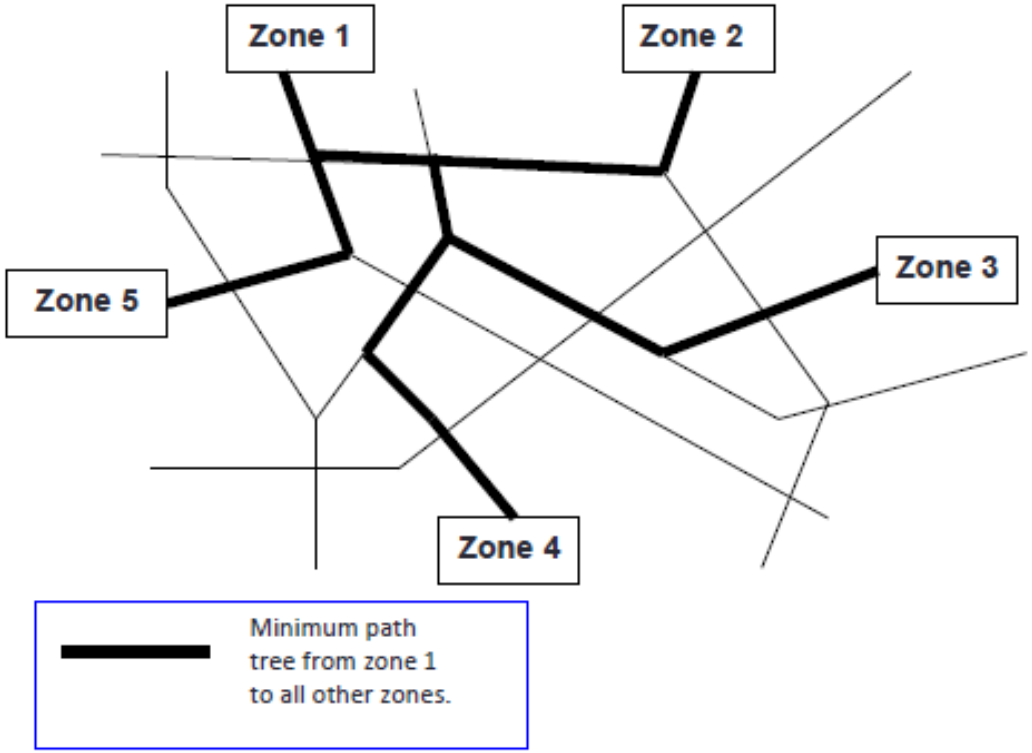
- Key step in Urban transport forecasting and planning.



Which Route will the people prefer?

  - Shortest route?
  - Fastest route?
  - Least costly route?
  - Scenic route?
- What happens if most cars choose a route?
- How to deal with congestion?

Trip Assignment (cont'd..)



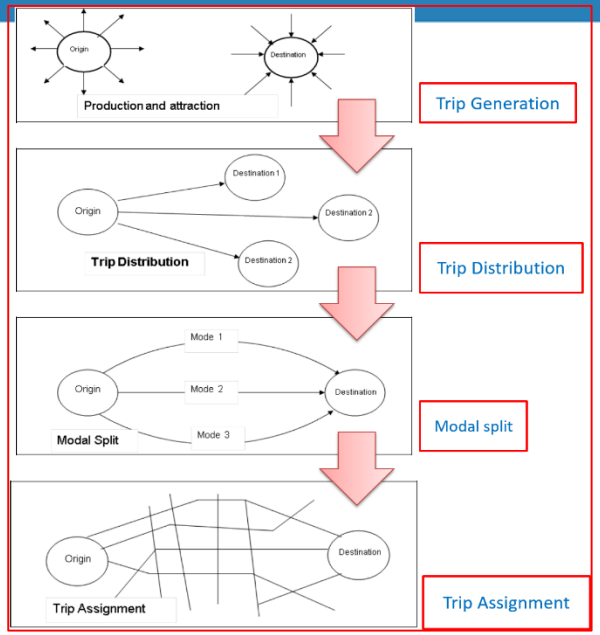
1. Estimates the traffic volume in each network.
2. Estimates travel costs for inter zones
3. Identifies bottle necks in the networks.

Source: (Riza Atiq bin O.K. Rahmat , 2012)

Critiques of the Traditional Planning Approach

- A tool created for a different goal. i.e. New road infrastructure.
- The Traditional Planning model "Predict and Accommodate" i.e. Forecasts based on peak spreading and Induced demand.
- Purely based on Quantitative numbers or Forecasted values.

Basic Approach remains unchanged even today, but is focused on System Management



Source: (MIT Lecture Notes, 2002)

“The McNamara Fallacy” in Traditional Planning Approach

“Measure whatever can be easily measured”.

implies……This is OK as far as it goes.

“Disregard that which cannot be measured easily.”

implies ……This is Artificial and misleading.

“Presume that which cannot be measured easily is not important.”

implies ……This is Blindness.

“Presume that which cannot be measured easily does not exist.”

implies ……This is Suicide.

Source: (https://expertprogrammanagement.com/2017/07/the-mcnamara-fallacy/)

Conclusion

- The four step Transport forecasting and planning method ensures selection of the most feasible transportation option for a selected area considering the local people needs and cost factors.

- Criticized for “Predict and Accommodate” approach and “Planning based on “Induced demand”.

System management should be addressed in parallel to the traditional transportation planning process to give emphasis on Accessibility rather than Mobility.

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Thank You